

Day 03 – Network interfaces

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THEORY

1. What is a Network Interface Card?

A Network Interface Card (NIC) is hardware that connects a device to a network. It can provide wired or wireless connectivity.

2. What is Ethernet?

Ethernet is a standard for wired network communication, commonly used in local area networks (LANs).

3. What is RJ-45?

RJ-45 is the common 8-contact connector used with Ethernet twisted-pair cables.

4. What does UTP stand for?

UTP stands for Unshielded Twisted Pair. It uses twisted copper wire pairs to reduce interference.

5. Name three common Ethernet cable categories.

UTP, single-mode fiber-optic, multimode fiber optic.

6. What is a straight-through cable?

A cable with the same wiring standard at both ends, commonly used to connect different types of network devices.

7. What is a crossover cable?

A cable with transmit and receive pairs crossed between its two ends, traditionally used to connect similar devices directly.

8. Explain Auto MDI-X.

Auto MDI-X automatically detects the required transmit and receive wiring, allowing devices to work with straight-through or crossover cables.

9. Difference between copper and fiber-optic networking?

Copper sends electrical signals through metal wires. Fiber sends data as light through optical fiber. Fiber supports longer distances and is resistant to electromagnetic interference.

10. What is multimode fiber?

Multimode fiber has a larger core and carries multiple light paths. It is mainly used for shorter distances and cheaper than single-mode fiber.

11. What is single-mode fiber?

Single-mode fiber has a small core and carries one main light path. It is used for long-distance, high-speed communication.

12. Two advantages of fiber-optic networking.

1. Supports long-distance communication.
- 2 No signals leaked outside of the cable so there is no security risk.

13. Why is understanding network interfaces important for cybersecurity?

It helps security professionals understand network connections, IP and MAC addresses, monitor traffic, configure firewalls, and identify suspicious activity.

PRACTICAL IDENTIFICATION

14. Identify your active network adapter.

Wireless LAN adapter Wi-Fi — Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC.

15. Write your IPv4 address, default gateway, and network adapter name.

IPv4 Address: 192.168.100.209

Default Gateway: 192.168.100.1

Network Adapter: Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC

SCREENSHOTS

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Windows PowerShell x Windows PowerShell + v
PS C:\Users\perehan_sewid> ipconfig /all

Windows IP Configuration

Host Name . . . . . : LAPTOP-AM22QKJ6
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-16
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::dd09:f6fe:ee7f:b4ac%22(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 789184551
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-55-B1-2C-00-FD-6B-09-4A-82
NetBIOS over Tcpip. . . . . : Enabled

Unknown adapter McAfee VPN:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TAP-Windows Adapter V9
Physical Address. . . . . : 00-FF-74-4B-36-3C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi 4:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC #4
Physical Address. . . . . : C6-13-75-3C-F1-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi 5:

Media State . . . . . : Media disconnected
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Windows PowerShell x Windows PowerShell + v

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC #4
Physical Address. . . . . : C6-13-75-3C-F1-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi 5:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC #5
Physical Address. . . . . : CA-13-75-3C-F1-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC
Physical Address. . . . . : C4-13-75-3C-F1-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::e3bb:c8ce:a438:c2e7%6(Preferred)
IPv4 Address. . . . . : 192.168.100.209(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Friday, September 4, 2026 10:31:42 PM
Lease Expires . . . . . : Tuesday, September 8, 2026 8:00:41 PM
Default Gateway . . . . . : 192.168.100.1
DHCP Server . . . . . : 192.168.100.1
DHCPv6 IAID . . . . . : 113513333
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-55-B1-2C-00-FD-6B-09-4A-82
DNS Servers . . . . . : fe80::1%6
                        8.8.8.8
                        8.8.4.4
                        fe80::1c97:c202:33cc:e623%6
                        fe80::1%6
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : C4-13-75-3C-F1-02
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
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